

INTRODUCTION

Welcome to the Supertails Data Analyst Assessment. This assignment is designed to evaluate your:

- SQL proficiency
- Python + Analytical thinking
- Dashboarding skills in Looker Studio
- Business intuition & insight storytelling

You are provided 5 datasets that mirror a real e-commerce + supply chain + communication ecosystem at Supertails.

Download dataset pack:

- orders.csv
- communication_logs.csv
- supply_chain.csv
- support_tickets.csv
- vet_calls.csv

Use these to complete the assignment.

DATA DICTIONARY

A short summary is included here for context.

1. orders.csv – 10,000 rows

Contains order-level details: order amount, category, city, promised vs actual delivery, payment mode, courier, etc.

2. communication_logs.csv – ~40,000 rows

Communication-level logs for each order across WhatsApp, SMS, Email, Call.

3. supply_chain.csv – 10,000 rows

Covers warehouse, scans, RTO, courier delays, weight, and TAT.

4. support_tickets.csv – 5,000 rows

Contains issue categories, resolution times, CSAT, escalations.

5. vet_calls.csv – 5,000 rows

Tracks vet consultation calls, transfer success, and call durations.

SECTION A — SQL ASSESSMENT

Write **clean, optimized SQL queries** for each question below.

A1. O2C & Communication Performance

1. Compute the **O2C connect rate** for WhatsApp messages (Clicked + Replied) ÷ Delivered.
2. Identify the **top 5 cities** with lowest O2C connect rate.

A2. Customer Purchase Behavior

1. Compute **repeat purchase rate** by
 - city
 - product_category
2. Build a **cohort table** for first-purchase month × repeat-purchase month.

A3. Delivery & Supply Chain

1. Calculate **promised vs actual delivery gap** in days for each order.
2. Identify **orders delayed due to courier_delay_flag = TRUE**.
3. Rank courier partners by **average shipment_tat_hours**.

A4. Communication Channel Insights

For each channel (WhatsApp/SMS/Email/Call), calculate:

- Delivery rate
- Read rate
- Click-through rate
- Reply rate

A5. Support Ticket Analysis

For each issue_category:

- Average resolution time (in hours)
- Escalation rate
- Average CSAT score

- Volume of tickets

A6. Vet Transfer Analysis

1. % of orders where a **vet consultation** happened within 72 hours of delivery.
2. Average duration (in minutes) of **successful vet transfers**.

SECTION B — PYTHON ANALYTICS (Jupyter Notebook)

Use **pandas, numpy, matplotlib/seaborn**.

B1. Data Cleaning

Perform and show:

- Missing value detection
- Removing invalid records
- Standardizing date formats
- Deduplicating any FK inconsistencies
- Validating order → supply chain timestamps

B2. Exploratory Data Analysis

Create **at least 7 visualizations** covering:

- Order value distribution
- Category-wise revenue
- City-level performance
- Payment mode breakdown
- Delivery TAT distribution
- Communication channel read/click effectiveness
- Support ticket trend by week

B3. RFM & Customer Segmentation

Using orders.csv:

- Build **Recency, Frequency, Monetary** scores
- Segment customers into **Gold / Silver / Bronze / At-Risk / New**
- Present insights supported by charts

B4. Predictive Analytics (Optional but Highly Preferred)

Build a simple model (Logistic Regression/Random Forest):

Target: Whether a customer raises a support ticket.

Features may include:

- Order value
- City
- Repeat status
- Shipment mode
- Communication engagement signals

Evaluate using AUC, accuracy, recall, F1.

B5. Supply Chain Insights

Using supply_chain.csv:

- Correlation analysis: **weight vs TAT vs shipment_mode**
- Identify top 10 **SKUs** with highest RTO probability
- Detect warehouse-level bottlenecks using hub_in_scan delays

SECTION C — LOOKER STUDIO DASHBOARD

Create a **fully interactive** dashboard with filters for date, city, courier, product category.

Mandatory Components

C1. Topline Metrics

- GMV
- Total Orders
- AOV
- Repeat Purchase %
- On-time delivery %
- RTO %

C2. End-to-End Sales Funnel

Orders → Shipped → Delivered → Returned → RTO

C3. Category & City Performance

Bar graph + heatmap

- Orders
- Revenue
- Cancellation %
- Delivery delays

C4. Communication Dashboard

By channel:

- Delivery %
- Read %
- Click %
- O2C connect %

C5. Supply Chain Dashboard

Visuals:

- Courier-wise TAT
- Warehouse performance
- Delay heatmap
- Mode comparison (Air vs Surface)

C6. Support & Vet Dashboard

- CSAT trends
- Issue category distribution
- Average resolution time
- Vet transfer success %

DELIVERABLES

Submit the following:

1. **SQL Queries (PDF or .sql)**
2. **Jupyter Notebook (.ipynb)**
3. **Looker Studio Dashboard Link**
4. **Insights Summary (2–3 pages max)**
5. **Optional:** PPT with business recommendations